Business Scope

STAGE 1

PriceWise: Pricing Intelligence & Decision-Support Dashboard

Executive Summary

PriceWise is an explainable pricing intelligence and decision-support dashboard designed to help business stakeholders evaluate product-level pricing decisions. Stage 1 focuses on the Enterprise segment and approximately 20–30 priority products.

The system uses recent product performance, historical trends, configurable business objectives, and pricing guardrails to provide a recommended price for stakeholder review.

PriceWise is a decision-support tool, not an autonomous pricing system. Price recommendations are refreshed daily, while stakeholders retain responsibility for accepting, rejecting, or editing a recommendation.

Business Problem

- Manual pricing review is time-consuming and difficult to scale across many products.
- Non-technical stakeholders who make pricing decisions may not have a clear, explainable way to evaluate recommended prices.
- Pricing decisions can be difficult to explain, compare, and validate over time.

PriceWise addresses this problem by providing a daily, explainable pricing decision-support framework. It combines rolling product performance, historical pricing context, configurable business objectives, and pricing guardrails to generate product-level price recommendations for stakeholder review.

Pricing Recommendation Framework

Stage 1 Scope

In Scope

- Enterprise segment only.
- Individual product-level pricing recommendations.
- Approximately 20–30 priority products.
- Historical monthly and fiscal-quarter performance analysis to track seasonality, end-of-quarter effects, and clearance pressure
- Revenue, units, margin, and clearance indicators.
- Configurable business objectives and weights.

- Explainable Daily Price recommendation refresh.
- Pricing guardrails.
- Human-in-the-loop accept, reject, and edit workflow.

Future Scope

- Personal and Partner segment pricing.
- Expansion beyond the initial 20-30 selected products.
- Demand-response and price-elasticity modeling.
- More rigorous profit optimization using validated cost and margin data.
- Inventory-aware optimization and clearance planning.
- Promotion, bundle, and product-mix effects.
- Cannibalization analysis across related products.
- Competitor price benchmarking if external data becomes available.
- Geographic or state-level pricing analysis.
- Automated pricing workflows and production-system integration.
- Cloud deployment and enterprise-scale architecture.
- Advanced conversational analytics.

Target Users & Stakeholders

- Pricing stakeholders: review and act on recommendations.
- Business/product stakeholders: understand commercial and product performance.
- Analytics/data stakeholders: validate data, metrics, and recommendation quality.
- Future pricing teams: potential users when additional segments are added.

Product-Level Scope

- Product-level metrics : Rolling Revenue, Rolling Units, Rolling ASP, Rolling Margin
- Family-level metrics : only as benchmark / comparison context
- Flag products with insufficient historical evidence rather than producing unreliable recommendations.
- Treat cannibalization among overlapping products as a future modeling consideration.

Product Selection

Stage 1 will focus on **20-30 high priority products for Segment 'Enterprise'.** These products are selected based on:

- Revenue contribution.
- Order-unit volume.
- Enterprise relevance.
- Availability of sufficient historical data.
- Potential pricing/business importance.

The final selection rule should be agreed with the stakeholder so that the portfolio can be reproduced consistently.

Pricing Decision Cadence

PriceWise will use the latest data to refresh product performance and generate a recommendation for stakeholder review, and the stakeholder decides whether a price change is appropriate.

- Daily: refresh available performance information and recommendations.
- Monthly: use aggregated data to identify broader performance patterns.
- Fiscal quarter: analyze seasonality, quarter-end effects, and clearance pressure.Fiscal Q1 begins in April, so fiscal periods should use the appropriate fiscal calendar. Add this part in the pricing formula

Data Cleaning & Business Rules

Duplicates

- No duplicates should be removed. Add a unique ID to each transaction.

Transaction Status

- Use Shipped/Ships transactions for pricing-performance calculations.
- Analyze returned transactions separately.
- Returned products may still be sellable inventory.

Missing Data

- The rows with missing product_type / product_name should be dropped completely
- Use order units instead of invoice quantity.
- Missing value return_date / reason_of_return / reason_of_cancel don't need to be dropped

Other Data Handling

- Tax is already included in revenue and does not need separate treatment for Stage 1.
- Rep_Take_Role is excluded from initial pricing logic.
- The raw synthesized dataset can be included in the project repository.
- Do not blindly remove outliers; investigate unusual Enterprise transactions, bulk orders, contracts, promotions, and potential data-quality issues first.

Core Business Metrics

The dashboard should provide a consistent product-performance view with metrics below. The word '***Rolling***' means the metric is recalculated every day using the most recent fixed time window, such as the latest 30 days or 90 days.

Metric	Stage 1 Calculation
Rolling Revenue (individual level/product family level)	Sum of shipped-invoice revenue by product family/individual product over the most recent rolling window
Rolling Units (individual level/product family level)	Sum of shipped-invoice order units by product family/individual product over the most recent rolling window
Rolling ASP (average sales price)	Rolling Revenue / Rolling Units
Rolling Margin (individual level/product family level)	Sum of shipped-invoice margin by product family/individual product over the most recent rolling window
Rolling Margin Rate	Rolling Margin / Rolling Revenue
Rolling Return Rate	Returned Units over rolling window / Shipped Units over rolling window
Sales Trend	Current rolling 30-day units - previous rolling 30-day units
Clearance Pressure	Negative sales trend or low rolling 30-day units relative to historical average
Price Change	Suggested Price - Current Reference Price

Objective Function

The objective function is used to translate product performance into a daily price suggestion. It combines historical pricing, recent product performance, and business priorities to decide whether the suggested price should move up, move down, or stay close to the baseline price.

Business Score Components

- PriceWise allows stakeholders to select the business objective that should receive greater emphasis when evaluating a potential price adjustment. Stage 1 supports four configurable objectives
- Revenue Protection: maintain commercially important products near a stable reference price while considering recent demand performance.
- Margin Improvement: favor an upward price adjustment when margin performance is below the stakeholder-defined target and demand remains sufficiently stable.
- Unit Movement: favor a downward adjustment when unit sales are declining and additional product movement is desired.
- Clearance Support: favor a downward adjustment when sustained weak sales or other available indicators suggest clearance pressure.

Daily Price Formula:	Suggested Price = Baseline Price * (1 + Adjustment Rate)
Baseline Price	w1 * *Current Rolling 30-Day Avg Price* + w2 * *Product Family Rolling 90-Day Avg Price* + w3 * *Previous Rolling 90-Day Avg Price* Default weights example : w1 = 0.5, w2 = 0.3, w3 = 0.2
Adjustment Rate	The adjustment rate is a directional pricing adjustment based on selected business signals. Potential adjustment factors can include: ● Business Score: revenue, margin, and unit performance. ● Sales Trend: recent increase or decrease in product sales. ● Clearance Pressure: whether weak recent movement suggests a need to lower price. ● Seasonality: fiscal-quarter or monthly pricing patterns. ● Stakeholder Objectives: user-selected priority weights.

Pricing Guardrails

Recommendations should be constrained by business rules to prevent unrealistic or operationally undesirable price changes.

- **Daily movement cap:** suggested price should not change by more than ±2% from the previous day's suggested price.
- **Monthly movement cap:** suggested price should not change by more than ±10% from the current/reference monthly average price.
- **Price-tier range:** suggested price should stay within the product's applicable price-tier range when reliable tier information is available.
- **Minimum margin floor:** suggested price should not fall below the minimum acceptable margin threshold once confirmed by stakeholders.
- **Data sufficiency flag:** products with insufficient historical data should be flagged for manual review rather than receiving an automatic recommendation.

Recommended Workflow

- Refresh and validate the latest transaction data daily.
- Filter to the Stage 1 scope: Enterprise segment, shipped-invoice records, and eligible products.
- Calculate rolling product metrics and pricing context.
- Apply the selected business-objective weights to generate the suggested price.
- Apply pricing guardrails and data-sufficiency checks.

- Present the recommendation with key explanation factors.
- Allow stakeholders to accept, reject, or override the recommendation.
- Record the final decision for future evaluation.

Dashboard Requirements

Tab 1 — Overview

- Enterprise pricing-performance summary.
- Revenue, units, margin, and trend indicators.
- Products requiring pricing attention.
- Top products by revenue and unit volume.
- Current/reference price versus recommended price.
- Filters for product, product type, family/series, fiscal period, and relevant business dimensions.
- Objective controls for changing business priorities.

Tab 2 — Product & Recommendation Detail

- Product name, type, and family/series.
- Current/reference price and recommended price.
- Percentage price change.
- Revenue, units, margin, margin rate, return-related indicators, and trends.
- Historical price/performance context.
- Objective contribution or score breakdown.
- Key factors explaining the recommendation.
- Accept, Reject, and Edit actions.
- Clear distinction between system recommendation and stakeholder-adjusted price.

15. Explainability Requirements

Each recommendation should be understandable in business terms.

- Identify the primary factors influencing the recommendation.
- Show recent revenue and unit performance.
- Show margin/profitability contribution.
- Show sales trend and clearance pressure.
- Show the effect of selected business objectives.
- Show the proposed distance from the current/reference price.
- Identify any guardrail that limited the recommendation.

What-If Analysis

Stakeholders should be able to test alternative business priorities without permanently changing the default configuration.

- Increase profitability/revenue/unit priority and observe the resulting recommendation.
- Increase clearance priority when inventory movement is more important.
- Enter custom objective weights.
- Show how the recommendation and supporting metrics change.

OPTIONAL – Chatbot (Explainable Assistant)

A conversational layer can help stakeholders query dashboard information and understand recommendations in natural language. Followed by the question 'How can I help you?', the chatbot will give several options. Selecting one option will switch to more detailed lists. Here are some options to consider:

Dashboard Overview

- What are the top revenue-driving products this month?
- Which products have the highest unit volume?
- Which products show declining sales performance?

Product Price History

- How has this product's price changed over recent periods?
- What is this product's current ASP compared with its product-family benchmark?

Recommendation Explanation

- Why is the recommended price higher or lower than the current/reference price?
- Which factors contributed most to this recommendation?
- Does this product have enough historical data for a reliable recommendation?

Objective Weights

- What are the default objective weights?
- What happens if I prioritize margin, revenue, units, or clearance?
- Let me specify my own objective weights.

Other Questions

- *This part is for personalized questions and will be developed in later stages.*

Human-in-the-Loop Decision Workflow

PriceWise supports human decision-making rather than automatically changing prices.

- Review the recommended price and key explanation factors.
- Accept, reject, or edit the recommendation.
- Record the final decision and override reason for future analysis.

Analytical Evaluation

Before relying on recommendations operationally, PriceWise should be evaluated using historical data.

- Back-test recommendations across multiple historical periods.
- Compare recommendations against current/reference prices and historical averages.
- Evaluate recommendation stability, guardrail frequency, and coverage across eligible products.
- If a demand model is added later, evaluate demand, revenue, and margin predictions separately.

Business Success Metrics

- Revenue impact.
- Margin / profit impact.
- Margin rate.
- Unit volume.
- ASP movement.
- Inventory reduction or clearance improvement.
- Recommendation acceptance rate.
- Stakeholder override rate.
- Recommendation coverage.
- Guardrail-trigger frequency.

Specific targets should be established after measuring a baseline with the client. Stage 1 should not assume a performance uplift without evidence.

Key Assumptions

- Enterprise is the initial target segment.
- Pricing suggestions are refreshed daily.
- Individual products are the primary pricing decision level.
- Product family / series is used for contextual benchmarking.
- Shipped transactions are the primary population for pricing-performance analysis.
- Stakeholders retain final authority over price changes.

Final Stage 1 Business Definition

PriceWise Stage 1 is an explainable, human-in-the-loop pricing decision-support solution for Lenovo's Enterprise segment. It provides daily product-level price recommendations for approximately 20-30 priority products using recent performance, historical context, configurable business objectives, and pricing guardrails.

The Stage 1 objective is to establish a transparent and testable pricing workflow that can be validated with stakeholders before expanding into advanced demand-based optimization or automated pricing capabilities.